A REMARK ON OPENAI FAMILY 196: THE ZERO-DIVISOR CONSTRUCTION WORKS WITH $q = 16$

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Abstract. OpenAI's manuscript *A Torsion-Free Group Algebra with Zero Divisors* (family 196 of `github.com/openai/math`, commit `adc7f124`) builds its graphs from the projective plane PG(2, 128): 16 513 ordinary letters, an alphabet of 16 516 letters, and a rose with 8 258 edges. We check every place where q enters the argument. The construction goes through unchanged for $q = 16$: 273 points, 276 letters, and a rose with 138 edges. The only nontrivial input is the squared-word lemma (Lemma 2.1). For $q = 16$ the manuscript's own weight vector certifies it with ratio $7138933/7179627 < 1$, and the manuscript's explicit $\delta = 1/600$ still holds. For $q = 8$ the lemma is false: the transfer matrix has spectral radius $1.01175 > 1$. Hence $q = 16$ is the smallest power of two for which the construction works.

1. Where q enters

We re-read all sections of the manuscript (`types.tex`, `entropy.tex`, `planar.tex`, `topology.tex`, `factors.tex`). The parameter q enters only at the following places.

(1) *Parity* (`types.tex` L28–31, `factors.tex` L139). Two lines meet in 1 or $q + 1$ points, and the extra parts meet in 0 or 2 letters. So $|S_x \cap S_y|$ is odd for all x, y if and only if q is even. Also $v - 3$ is even for every q.

(2) *Squared words* (Lemma 2.1, L72–101). The lemma requires the spectral radius of $M_{t,u} = w(t, u)^2$ to be less than 1. It is the only input to the entropy section that depends on q: `entropy.tex` uses only $\delta > 0$, the fixed alphabet $|T|$, and the least positive weight, through a_0.

(3) *Minimum degree.* The short-closure lemma (`planar.tex` L63) needs $(q + 1) - 2 \geq 3$. The expansion count in Lemma 2.4 (`types.tex` L158–167) needs $r/k \geq (q + 1)/2 > 2$; for $q = 128$ this is the exponent $62.5 = 64.5 - 2$.

(4) *Existence of constants.* The constant c in (2.5), and the constants D_0, d, D, ε that follow from it, depend on q, but only their existence is used. With $q = 16$ the value $c = 1/100$ still works: $c \log(2|T|/p) = 0.091 < 1$ and $2c \log d_* = 0.060 < 1$, where $d_* = q + 4 = 20$.

The remaining numbers in Section 2.1 only illustrate $q = 128$. These are the counts at $n = 2v^2$, the value $16528 < \mathrm{e}^{10}$, the weight range $[1/129, 1]$, and $\delta = 1/600$, $h \geq 3002$. For $q = 16$ they become: at $n = 2v^2 = 149\,058$, each line class on A has 17 vertices of each double-extra type and 495 of empty type, and on B it has 34 triple-extra and 512 empty; $\sum f = 288$; the weights lie in $[1/17, 1]$; and $\delta = 1/600$ holds for $h \geq 2412$.

2. The squared-word lemma for $q = 16$

Classify letters as O_0 ($t, \bar{t}$ ordinary), O_1 ($\bar{t}$ extra, three letters) and E (t extra, three letters). The weight $w(t, u)$ depends only on the classes of $\bar{t}$ and u. Hence this partition is equitable for

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M, with quotient

$$Q = \begin{pmatrix} (v-4)o & 3o & 3e \\ (v-3)e & 3e & 2 \\ (v-3)o & 2o & 3e \end{pmatrix}, \qquad o = \frac{1}{(q+1)^2},\ e = p^2,\ p = \frac{q+1}{v}.$$

The manuscript's f is constant on classes ($f = 5$ on O_1, $f = 1$ otherwise). So $(Mf)(t)/f(t)$ takes exactly three values:

$$\frac{q}{q+1} + \frac{12}{(q+1)^2} + 3p^2, \qquad \frac{2 + (v+12)p^2}{5}, \qquad \frac{v+7}{(q+1)^2} + 3p^2.$$

The first two are exactly the row bounds of the manuscript (L83, L89). At $q = 128$ they equal 500731261911/504183783481 and 820350863/1363395845, the rationals printed there.

Proposition 1. *For $q = 16$, $Mf \le \lambda_{16} f$ with $\lambda_{16} = 7138933/7179627 = 0.99433\ldots$ (attained on O_0; the other two values are 77141/124215 and 7039561/7179627). Consequently $\sum_{W \in T^h} P(W)^2 \le 288\, \lambda_{16}^{h-1} \le \mathrm{e}^{-h/300}$ for $h \ge 2412$, so Lemma 2.1 holds with $\delta = 1/600$. The exact Perron root is $\rho(M) = 0.975008$, the largest root of $x^3 - \frac{6849809}{7179627}x^2 - \frac{2458920281}{178363473561}x - \frac{215608591}{33320648907}$.*

Proposition 2. *For $q = 8$, $\rho(M) = 1.011749$. A rational positive class-constant vector g gives $\min_t (Mg)(t)/g(t) = 70905723398671/70082342865504 > 1$, so by the Collatz–Wielandt inequality $\rho(M) > 1.0117$. Since $\sum_{W \in T^h} P(W)^2 = \mathbf{1}^{\mathsf{T}} M^{h-1} \mathbf{1} \ge c\, \rho(M)^{h-1}$ for some $c > 0$, Lemma 2.1 fails for every $\delta > 0$. The same holds for $q = 2$ ($\rho = 1.6238$) and $q = 4$ ($\rho = 1.1825$).*

3. Verification

`scripts/independent/kz196_indep.py` builds $\mathrm{PG}(2, q)$ over $\mathrm{GF}(q)$ in Sage for $q \in \{2, 4, 8, 16, 32\}$. From the plane it builds the explicit vertex types at $n = 2v^2$ and checks: integrality of the counts, letter frequency p on both sides, and odd cross-intersections for all type pairs (exhaustively for $q \le 16$). It then computes $w(t, u)$ *empirically* from the B-side type counts, checks that the A-side frequencies are at most w, checks that the three-class partition is equitable and that its quotient equals Q, and finally computes the exact certificates above. Numerical eigenvalues of the full matrix agree with the exact Perron root. The primary scripts (`kz196_exact.py`, `kz196_spectral.py`) take the weights from the formula instead of the geometry and give the same numbers.

Remark 3 (Other orders). $\rho(M)$ depends only on q, so any projective plane of even order q could be used. The formula gives $\rho < 1$ already for $q = 10, 12, 14$ (0.9912, 0.9816, 0.9770). There is no plane of order 10 (Lam et al.), and the existence of planes of order 12 or 14 is open. So $q = 16$ is the smallest order for which a plane is known to exist and the construction works. Among Desarguesian planes $\mathrm{PG}(2, q)$, which the manuscript uses, it is the smallest.

4. What this does and does not change

The rose, and hence the generating set of G, shrinks from 8 258 to 138 generators, and the vertex degrees drop from 129–132 to 17–20. The group G itself still comes from a random graph on $2n$ vertices with $n \to \infty$ unspecified, and it has one cone (relator family) per graph component. This remark therefore does not make the counterexample explicit or computer-checkable, and it does not bound the number of relators. It also says nothing about other type designs. For $q = 8$, a different choice of extra letters might give a contracting transfer matrix; we have not explored this. The Lean formalisation hard-codes $q = 128$ (for example `TypedDiameter.lean`, `WordEstimate.lean`) and was not modified.

References

[1] OpenAI, *A Torsion-Free Group Algebra with Zero Divisors*, September 23, 2026, `https://github.com/openai/math/tree/adc7f1241b42e322a6451854ab7e4b4c146bf78a/preprints/A-Torsion-Free-Group-Algebra-with-Zero-Divisors-September-23-2026`.

[2] C. W. H. Lam, L. Thiel, S. Swiercz, The non-existence of finite projective planes of order 10, *Canad. J. Math.* 41 (1989), 1117–1123.